9. The following data gives the average life(in hours) and range of 12 samples of 5lamps each. The data are

Sample No	1	2	3	4	5	6
Sample Mean	1080	1390	1460	1380	1230	1370
Sample Range	410	670	180	320	690	450
Sample No	7	8	9	10	11	12
Sample Mean	1310	1630	1580	1510	1270	1200
Sample Range	380	350	270	660	440	310

Construct control charts for mean and range. Comment on the control limits.

10. The following are the sample means and ranges for 10 samples, each of size 5. Calculate the control limits for the mean chart and range chart and state whether the process is in control or not.

Sample number	1	2	3	4	5	6	7	8	9	10
Mean	5.10	4.98	5.02	4.96	4.96	5.04	4.94	4.92	4.92	4.98
Range	0.3	0.4	0.2	0.4	0.1	0.1	0.8	0.5	0.3	0.5

Table of Control Chart Constants

Sample Size	A2	D3	D4
2	1.880	0	3.267
3	1.023	0	2.574
4	0.729	0	2.282
5	0.577	0	2.114
6	0.483	0	2.004
7	0.419	0.076	1.924
8	0.373	0.136	1.864
9	0.337	0.184	1.816
10	0.308	0.223	1.777
11	0.285	0.256	1.744
12	0.266	0.283	1.717
13	0.249	0.307	1.693
14	0.235	0.328	1.672
15	0.223	0.347	1.653
16	0.212	0.363	1.637
17	0.203	0.378	1.622

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